



ChemBuddy

The Intelligent Academic Operating System for MSc Chemistry Students

Version 3.5.2 • Complete Visual Operating Manual

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Target Curriculum: MSc Organic, Inorganic, Physical & Analytical Chemistry

University Alignment: BCU, Central Universities & UGC Model Syllabus

Platform: Android (Flutter SDK • 100% Offline Core • Gemini AI)

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1. Introduction & Academic Architecture

ChemBuddy is an advanced, offline-first mobile application designed specifically to eliminate the academic friction experienced by Master of Science (MSc) Chemistry postgraduates. Developed by **Prajwal A Kambar**, it merges rigorous chemoinformatics, high-precision spectroscopy solvers, attendance predictive mathematics, and state-of-the-art generative AI into a unified laboratory and classroom companion.

Core Engineering Principles

- **100% Offline Core:** Attendance calculations, timetable intelligence, 16 postgrad calculators, Woodward-Hoffmann pericyclic selection rules, spectroscopy databases, and reaction prediction algorithms function entirely on-device without internet connectivity.
- **Scientific Rigor:** All formulas, mechanisms, spectroscopic chemical shifts, and thermodynamic equations strictly adhere to IUPAC standards and standard graduate textbooks (Silverstein, Pavia, March, Clayden).
- **Student Privacy & Local Ownership:** Student profiles, attendance logs, notes, and uploaded PDFs reside securely in encrypted local Hive boxes on your device. Zero telemetry or surveillance.
- **Dark Laboratory Ergonomics:** The UI utilizes high-contrast deep purple (#0A0914) and cyan (#06B6D4) palettes engineered to preserve dark-adapted vision in laser labs and darkrooms.

2. Installation, Android Permissions & Initial Setup

1. **Download Release APK:** Download `ChemBuddy-v3.5.2-by-Prajwal-A-Kambar.apk` from the official GitHub Releases page.
2. **Enable Unknown Sources:** In Android Settings, navigate to *Security* → *Install Unknown Apps* and allow your browser or file manager to install APKs.
3. **Grant Essential Permissions:**
 - **Notifications:** Required for the 8:30 AM Timetable Briefing, 30-min class alerts, and 8:00 PM Psychology Facts.
 - **Alarms & Reminders (Android 12+):** Required so that scheduled class notifications fire with exact timing.
 - **Battery Optimization:** Set to "Unrestricted" so Android Doze does not suppress morning notifications.
4. **Profile Initialization:** On first launch, enter your Full Name, Register Number, College/University, Semester, and Specialization (Organic, Inorganic, Physical, or Analytical). Your name is permanently remembered and used across the app and export dossiers.

3. 🌟 Ask ChemBuddy AI — The Chemistry Intelligence Engine (FLAGSHIP FEATURE)

💡 Flagship Feature Spotlight: ChemBuddy AI

ChemBuddy AI is your 24/7 MSc Chemistry Research & Exam Companion. Accessible via the **Ask AI** bottom navigation tab or the Home Screen action card, it connects directly to Google Gemini models (with intelligent offline fallbacks) and provides 9 specialized academic modes, chemistry-aware speech recognition, and instant active recall flashcard generation.

3.1 Overview of the ChemBuddy AI Interface

The Ask ChemBuddy screen is designed to behave like an interactive academic interview with a senior chemistry professor. Key components include:

- **Horizontal Mode Scroller:** Tap between 9 specialized AI modes instantly.
- **Chemistry Speech-to-Text:** Tap the pulsing purple microphone button to dictate questions verbally. ChemBuddy automatically normalizes spoken phrases (e.g. converting "SN1" to S_N1 , "Huckel rule" to $4n+2$, and "H-NMR" to $^1H\text{ NMR}$).
- **KaTeX Equation Box:** Mathematical and chemical expressions are formatted in crisp, textbook LaTeX notation.
- **Electron Pushing & Curved Arrows:** Step-by-step mechanism breakdowns detail electron displacement from nucleophile to electrophile.
- **1-Tap Flashcard Generation:** Save any AI response directly into your spaced repetition flashcard deck with a single tap.

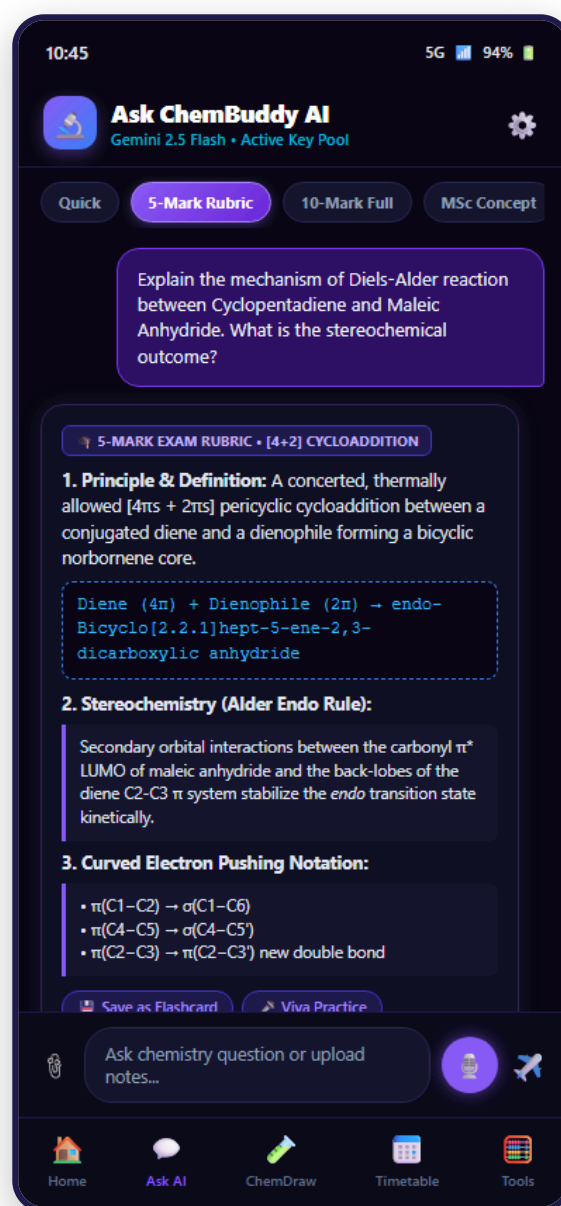


Figure 3.1: Ask ChemBuddy AI Interface with 5-Mark Exam Rubric mode, KaTeX math, and step-by-step electron pushing notation.

3.2 The 9 Specialized AI Modes Explained

Mode Name	Target Output	Best Used For
1. Quick Answer	High-yield, concise fact	Quick lab lookups, boiling points, reagents, oxidation states, and formulas.
2. General AI	Broad generative reasoning	Open-ended chemistry discussions, general scientific writing, and multi-turn research inquiries.
3. Simple Explanation	ELI5, jargon-free analogies	Breaking down intimidating concepts (e.g., Berry pseudorotation, NMR relaxation times) into plain English.
4. 2-Mark University Exam	Concise, high-density definition	Direct definitions, IUPAC rules, and single-equation university exam questions.
5. 5-Mark Exam Rubric	Structured 5-section essay	Standard university exam questions: 1) Principle, 2) Balanced Reaction, 3) Mechanism, 4) Scope, 5) Summary.
6. 10-Mark Full Rubric	Comprehensive long-form essay	Major university questions with curved-arrow pushing, orbital transition states, and stereochemical proofs.
7. MSc Concept Deep-Dive	Orbital & thermodynamic theory	Frontier Molecular Orbital (FMO) analysis, perturbation theory, and physical organic chemistry intuition.
8. Stepwise Mechanisms	Curved arrow electron flow	Detailed step-by-step reaction mechanisms, nucleophilic attacks, carbocation rearrangements, and driving forces.
9. From My PDF (RAG)	Strictly grounded in notes	Answers locked 100% to your uploaded syllabus, class notes, or research papers. Zero hallucinations.

3.3 Chemistry Speech-to-Text Normalization

When dictating via the microphone button (), ChemBuddy passes speech through the proprietary `ChemistryTextFormatter.normalizeSpeechQuery()` engine. Common spoken terms are converted to proper scientific formatting:

```
"s n 1 reaction" → "S_N1 reaction"
"huckel 4 n plus 2 rule" → "Hückel (4n+2) π-electron rule"
"proton n m r splitting" → "1H NMR splitting pattern"
"diels alder reaction" → "Diels-Alder [4+2] cycloaddition"
```

3.4 Multi-Key Gemini AI Orchestrator

To guarantee 99.9% uptime and prevent Google Gemini rate limits (HTTP 429), ChemBuddy utilizes an intelligent key-pool orchestrator:

- **Round-Robin Key Rotation:** Distributes requests across multiple API keys.

- **Automatic Failover:** If a key reaches its quota, the app automatically fails over to the next key without alerting the user.
- **Model Fallback Hierarchy:** Automatically tries `gemini-2.5-flash` → `gemini-2.0-flash` → `gemini-1.5-flash`.
- **Offline Synthesizer:** If completely offline, the local reaction engine and textbook databases take over seamlessly.

4. Home Dashboard, Decluttered Header & Personalized Greetings

The **Home Dashboard** serves as the central command center of ChemBuddy. It gives you an immediate overview of your academic health, daily schedule, and cognitive enrichment upon opening the app.

4.1 Decluttered Ergonomic Header (v3.5.2)

In v3.5.2, the Home screen header has been streamlined from a bulky 250px multi-line layout to a compact, punchy **65px single-line header**:

- **Personalized Student Name:** Replaced the generic "Chemist" placeholder with your actual name entered during login or registration (e.g. "Good morning, Prajwal 🧠" or "In the flow, Prajwal 📖").
- **Single-Line Truncation:** Font size reduced to crisp 17px with single-line overflow ellipsis, eliminating ugly line-wrapping.
- **Compact Subtitles:** University details and daily chemical quotes formatted into elegant, unobtrusive single-line tags.

4.2 Attendance Health Ring

A circular progress ring visually communicates your university compliance status:

- **Green ($\geq 75\%$):** Exam Eligible. Displays the exact number of classes you can safely skip.
- **Red ($< 75\%$):** Attendance Shortage. Displays the exact consecutive classes needed to recover eligibility.

4.3 Daily Psychology Facts Engine

Inspired by the popular "Psychology Facts" app, ChemBuddy includes a curated database of **120+ evidence-based cognitive insights**. A new fact rotates daily at 8:00 PM IST using a prime-weighted calendar algorithm.



Figure 4.1: Home Dashboard showing the decluttered greeting with personalized name, attendance health ring (82.4%), and daily psychology fact.

5. Smart Timetable & Attendance Intelligence Engine

The attendance module uses predictive mathematical equations to keep students compliant with the university 75.0% threshold.

5.1 The 75% Cutoff Mathematics

1. Classes Needed to Recover 75%:

$$\text{Needed} = \max\left(0, \lceil 3 \times (\text{Absent} - \text{Present}) \rceil\right)$$

Indicates the exact consecutive classes required without a single absence to reach 75.0%.

2. Safe Bunks Available:

$$\text{Can Skip} = \max\left(0, \left\lfloor \frac{(\text{Present} - 3 \times \text{Absent})}{3} \right\rfloor\right)$$

Shows how many classes can be safely skipped while remaining strictly above 75.0%.

5.2 Attendance States

- **Present (✓):** Increments both Present and Total counts.
- **Absent (X):** Increments Total count only.
- **Postponed / Cancelled (||): Excluded from calculations.**
Does not penalize your percentage when a professor is absent or a holiday occurs.
- **On Duty (OD):** Treated as Present for official university events.

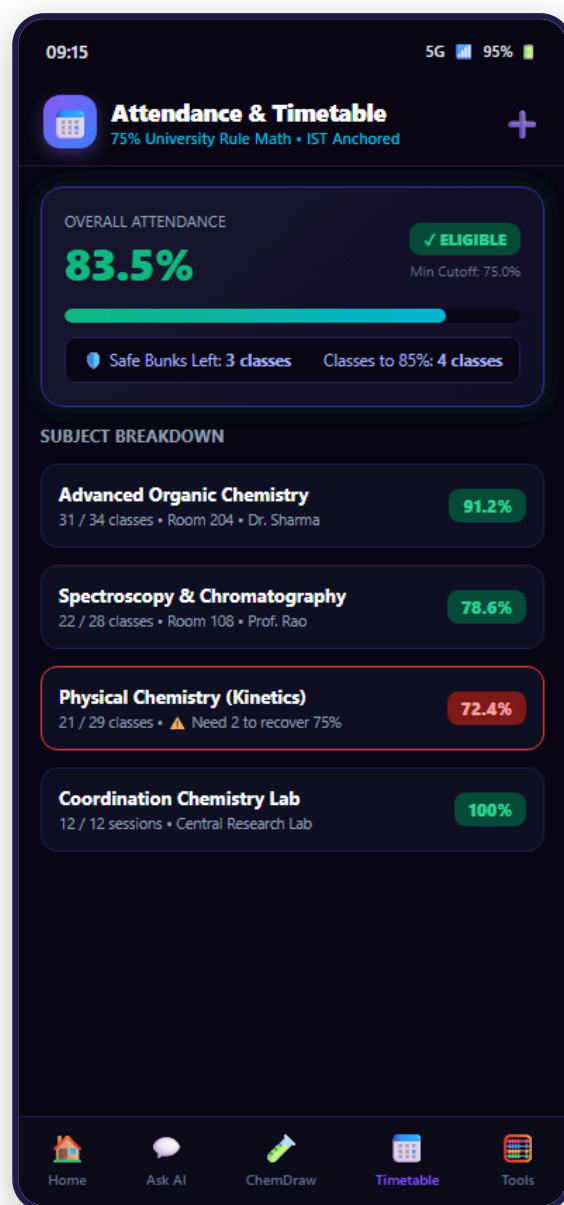


Figure 5.1: Timetable & Attendance Screen displaying the overall 83.5% meter, safe bunk calculation, and per-subject breakdown.

6. Notification System & 1-Tap Action Shade

ChemBuddy includes a specialized notification engine anchored strictly to the **Asia/Kolkata (IST)** timezone:

6.1 1-Tap Action Shade

When a class notification arrives 30 minutes before your lecture, you don't even need to open the app! Directly in the Android notification shade, tap:

- **Present ✓**: Instantly logs your attendance as present in the local database.
- **Absent ✗**: Instantly logs your attendance as absent.

6.2 Notification Channels

- **8:30 AM Timetable Briefing**: Daily morning overview of all scheduled lectures, room numbers, and attendance health.
- **30-Min Class Alerts**: Pre-class reminder with room number and current subject attendance.
- **8:00 PM Psychology Facts**: Daily cognitive insight to foster productive study habits.

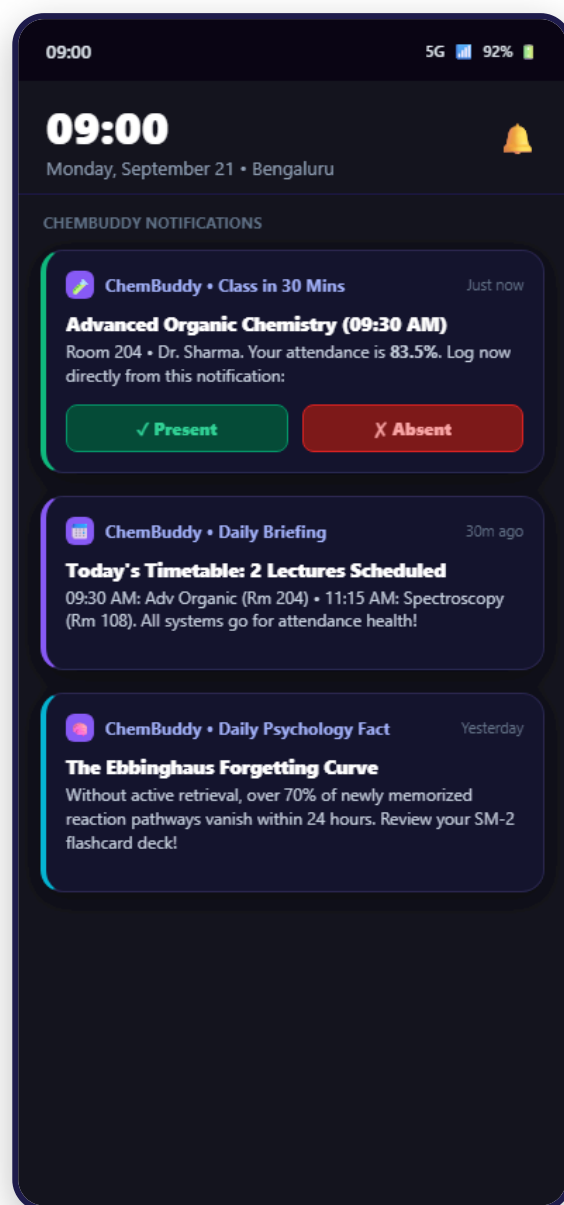


Figure 6.1: Android Notification Shade showing 1-tap Present/Absent action buttons, 8:30 AM briefing, and 8:00 PM psychology fact.

7. ChemDraw Pro & Ketcher Pro (118-Element Molecular Sketcher)

8. Reaction Product Prediction & 2D Vector SVG Generator

7.1 Molecular Sketcher Capabilities

- **Full 118-Element Periodic Table:** Searchable modal with standard CPK color coding.
- **Bond Tools:** Single, double, triple, wedge (stereochemical up), dash (stereochemical down), and curved electron arrows ($\curvearrowright$).
- **Formal Charges:** +1, -1, +2, -2 with circled badges and M CHG Molfile export.
- **Marquee Selection ($\square$) & 2D Clean-Up (🧹):** Standardizes bond lengths to 55px and optimizes angles.
- **Multi-Reactant Mode:** When two disconnected molecules are drawn on canvas, ChemBuddy renders a cyan + symbol between them and outputs dot-separated SMILES (e.g. C1=CCC=C1.c1ccccc1).

8.1 Reaction Predictor & 2D Vector SVG

- **Gemini AI Internet Prediction:** Queries Google Gemini AI to predict the major product, formal reaction name, mechanism class, and curved electron-pushing arrows.
- **Publication-Quality 2D Vector SVG:** Replaced raw text boxes with crisp, scalable vector SVGs rendered directly in the prediction bottom sheet for Diels-Alder adducts, biaryls, and rings.
- **45+ Instant Offline Rules:** Zero-latency local fallback ensures predictions always succeed even without internet.

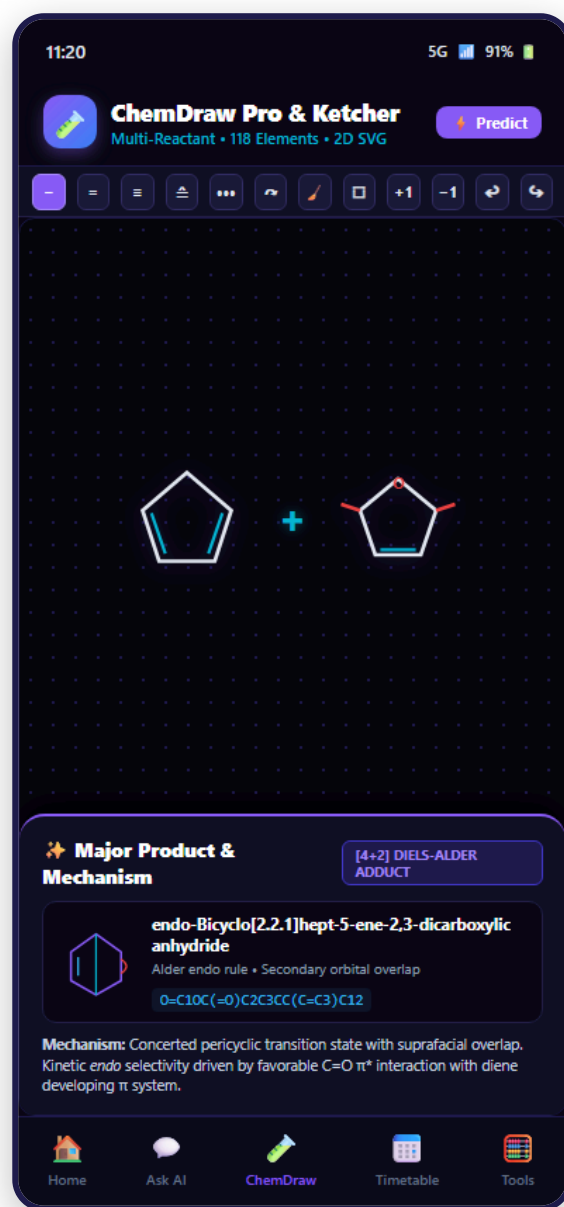


Figure 8.1: ChemDraw Pro canvas with Cyclopentadiene + Maleic Anhydride and the 2D Vector SVG prediction bottom sheet.

9. 🌩️ Reaction Mechanism Dossier & Compendium Export (PDF, Excel/XLV & SVG)

🌟 Publication-Grade Reaction Export Engine (v3.5.2)

ChemBuddy v3.5.2 introduces a complete, professional reaction export engine. Students can now export any individual reaction mechanism or the entire 21+ MSc Reaction Compendium into publication-grade Academic PDFs, Excel-compatible CSV spreadsheets (XLV format), and standalone Vector SVG diagrams with a single tap.

9.1 Academic PDF Dossier Features

Exported PDF dossiers adhere strictly to university publication formatting (0.5 inch margins, deep purple `#4C1D95` header bars, and Helvetica typography):

- **Reaction Scheme & Vector Diagram Block:** Visual equation banner displaying `Reactants → [Reagents / Conditions] → Products` with vector SVG diagram annotations.
- **Detailed Step-by-Step Mechanisms:** Full breakdown of each elementary step:
 - **Step Number & Title:** (e.g. *Step 1: Protonation of Carbonyl Oxygen*)
 - **Chemical Transformation:** In-depth explanation of orbital overlap, leaving group departure, and resonance stabilization.
 - **Curved Arrow Electron Flow:** Precise notation of electron movement (e.g. *"Lone pair on oxygen attacks carbonyl carbon, π electrons shift to oxygen"*).
 - **Intermediates & Reactive Species:** Molecular formula and geometry of intermediate carbocations, enolates, or transition states.
- **Multi-Page Pagination:** Uses Syncfusion PDF pagination so that complex 4-to-5 step mechanisms flow across pages naturally with zero text clipping or overlapping.
- **Synthetic Applications & Limitations:** Bulleted lists detailing pharmaceutical/industrial utility, regioselectivity, stereochemistry, and side reactions.
- **Academic Footer:** Verified footer on every page: *"Chem Buddy by Prajwal A Kambar - MSc Chemistry Reaction Dossier"*.

9.2 Excel Spreadsheet Export (CSV / XLV Format)

All spreadsheets are generated with a **UTF-8 Byte Order Mark (BOM)**, allowing Microsoft Excel and Apple Numbers to open the file cleanly without character encoding corruption:

- **Single Reaction Monograph:** Contains reaction metadata headers, followed by a structured **Step-by-Step Mechanism Table** with columns: `Step Number`, `Step Title`, `Chemical Transformation`, `Curved Arrow Movement (Electron Flow)`, and `Intermediate / Reactive Species`. Includes the full SVG vector code block!

- **Complete MSc Compendium (Catalog):** Comprehensive dataset with columns: Reaction ID , Reaction Name , Category , Aliases , Reactants , Reagents & Conditions , Products , Regioselectivity , Stereochemistry , Driving Force , Step Count , Detailed Step-by-Step Mechanism , SVG Vector Diagram Code , Key Applications , and Limitations .

9.3 Standalone Vector SVG Diagram Export

Tap "**Vector SVG Diagram (.svg)**" to export the pure, standalone vector graphics file. This `.svg` file can be opened directly in vector design software (Adobe Illustrator, Inkscape, ChemDraw), inserted into Microsoft Word or PowerPoint for seminar presentations, or embedded in web pages with infinite scalability.

9.4 How to Export Reactions in ChemBuddy

1. From Reaction Mechanisms Screen:

- **Single Reaction:** Tap any reaction from the list → tap the **Share / Export icon** (🔗) in the top-right corner → select *Academic PDF Dossier*, *Excel Spreadsheet*, or *Vector SVG Diagram*.
- **Complete Compendium:** On the main reactions catalog → tap the **Download icon** (📥) in the top-right corner → select *Complete Compendium PDF* or *Catalog Spreadsheet*.

2. From Profile Screen:

Scroll down to "Academic Exports & Dossiers" → tap "Export Reaction Mechanisms".

10. Spectroscopy & Chromatography Hub & Structure Solver

The **Spectroscopy Hub** covers 8 major analytical techniques and features an intelligent Structure Solver capable of identifying unknown compounds from spectroscopic data.

10.1 8 Analytical Techniques

- **RP-HPLC & GC:** Retention times, plate counts (N), resolution (Rs).
- **500 MHz ^1H NMR & ^{13}C DEPT:** Chemical shifts, multiplicity, coupling constants (J), DEPT-45/90/135.
- **FT-IR:** Characteristic functional group frequencies (carbonyls, O-H, N-H, triple bonds).
- **Mass Spectrometry:** Molecular ion peak (M^+), base peak, McLafferty rearrangement.
- **UV-Vis:** Woodward-Fieser rules for conjugated dienes and enones.

10.2 Structure Solver Intelligence (52+ Molecules)

- **Double Bond Equivalent (DBE):** Automatically computes degrees of unsaturation from molecular formula.
- **Constitutional Isomer Ruling-Out:** Explains systematically why alternative isomers are excluded based on coupling patterns and chemical shifts.
- **Interactive Zoom (1.0x - 5.0x):** Smooth pan and zoom with haptic peak snapping.

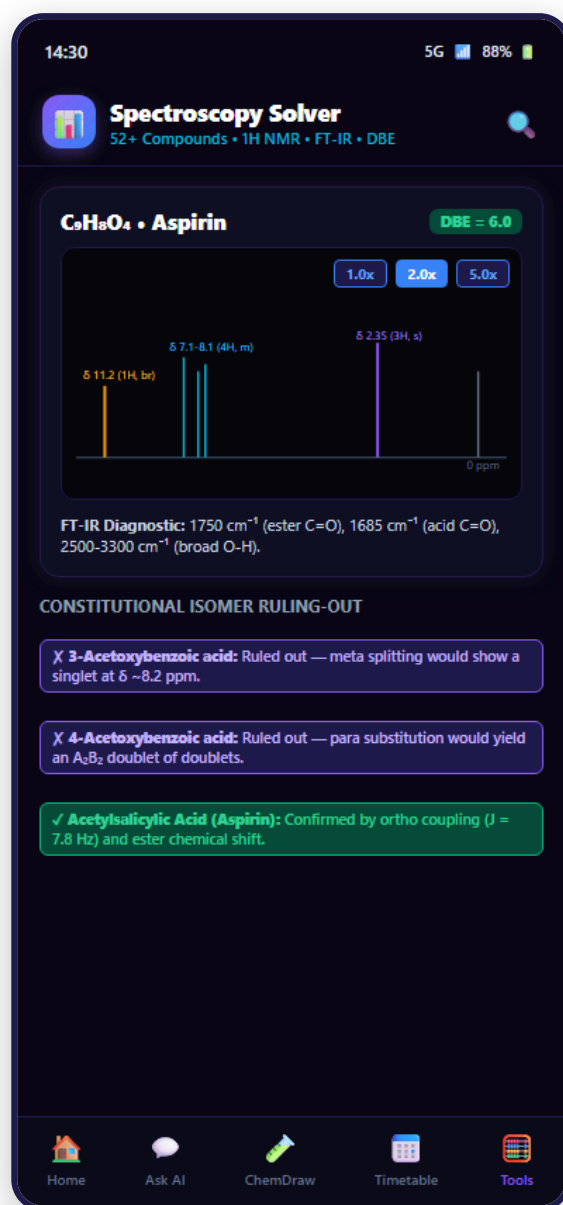


Figure 10.1: Spectroscopy Solver analyzing Aspirin ($\text{C}_9\text{H}_8\text{O}_4$, DBE=6.0) with zoomable ^1H NMR spectrum and isomer exclusion rationale.

11. MSc Chemistry Toolkit & 16 Postgrad Calculators

ChemBuddy includes 16 specialized postgrad calculators organized into 6 practical laboratory domains:

- **Solutions & Concentrations:** Molar Mass, Molarity (g/L & M), Dilution ($C_1 \cdot V_1 = C_2 \cdot V_2$).
- **Acid-Base & Buffers:** pH/pOH, Henderson-Hasselbalch, Buffer Formulation Assistant (Acetate, Phosphate).

- **Thermodynamics & Kinetics:** Gibbs Free Energy ($\Delta G = \Delta H - T\Delta S$), Arrhenius Rate Law (Activation Energy E_a).
- **Spectroscopy:** Beer-Lambert Law ($A = \epsilon \cdot c \cdot l$), Photon Energy ($E = h \cdot c / \lambda$).
- **Electrochemistry:** Nernst Equation ($E = E^\circ - (RT/nF) \cdot \ln(Q)$), Standard Cell Potential.
- **Chromatography:** HPLC Calibration Curve ($y = mx + c$), Theoretical Plates ($N = 16 \cdot (t_R/W)^2$), Resolution (R_s), Kovats Retention Index (I).

12. Pericyclic Hub & Woodward-Hoffmann FMO Engine

- **Frontier Molecular Orbital (FMO) Theory:** Visualizes HOMO and LUMO symmetry under thermal (Δ) and photochemical ($h\nu$) conditions.
- **Electrocyclic Reactions:** $4n$ systems (Thermal: Conrotatory, Photo: Disrotatory) and $4n+2$ systems (Thermal: Disrotatory, Photo: Conrotatory).
- **Cycloadditions:** $[4+2]$ Diels-Alder (Suprafacial-Suprafacial) and $[2+2]$ Photocycloadditions.
- **Sigmatropic Rearrangements:** $[1,3]$, $[1,5]$, and $[3,3]$ Cope and Claisen rearrangements.

13. Smart Flashcards & Spaced Repetition (Active Recall)

The **Smart Flashcards** engine implements the scientifically proven **SuperMemo-2 (SM-2)** spaced repetition algorithm to guarantee long-term retention of reaction mechanisms, reagents, and spectroscopic values.

13.1 4-Point Confidence Ratings

After flipping a card to reveal the answer, rate your recall:

- **Again (1):** Interval resets to <1 day.
- **Hard (2):** Modest interval increase (~3 days).
- **Good (3):** Standard exponential increase (~6 days).
- **Easy (4):** Maximum interval increase (~12+ days).

13.2 Auto-Deck Generation

Tap "**Save as Flashcard**" on any response inside Ask ChemBuddy AI or import a syllabus PDF to automatically generate a tailored flashcard deck with questions and model answers.

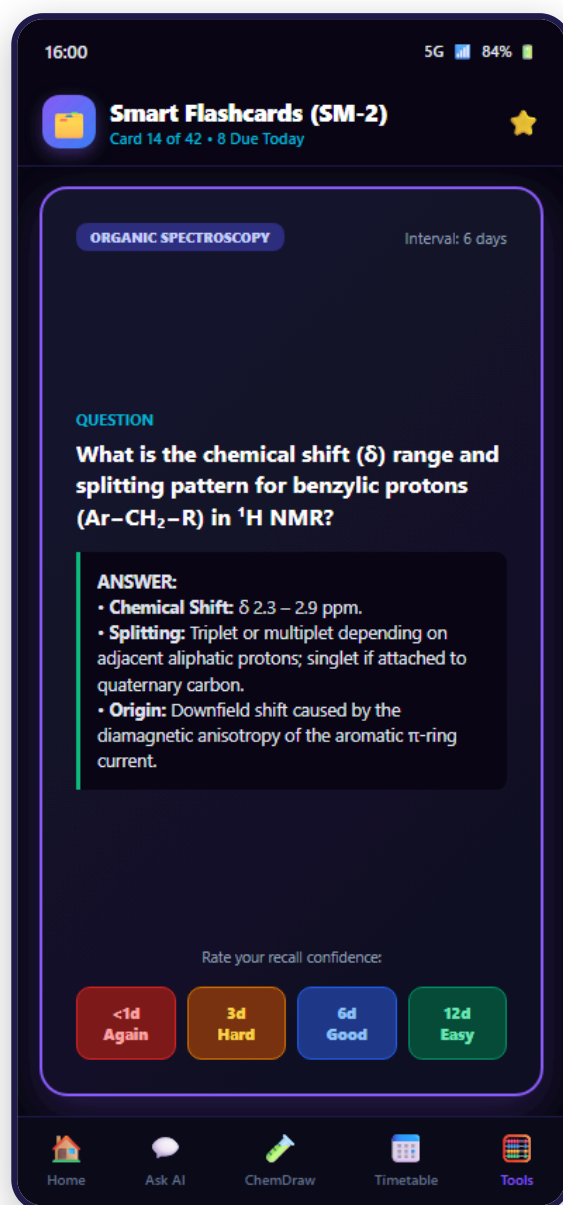


Figure 13.1: Flashcards active recall screen showing question card, diagnostic answer breakdown, and SM-2 rating buttons.

14. PDF Library, Study Hub & Strictly Grounded RAG Assistant

- **Offline PDF Viewer:** High-performance document reader with page jumping, keyword search, and zoom.
- **Local Text Extraction:** Extracts text locally on your device without uploading files to third-party servers.
- **Strictly Grounded RAG Chat:** In "From My PDF" mode, ChemBuddy's AI responses are strictly grounded in your uploaded documents. If the answer is not in the text, it explicitly states so, preventing hallucinations.

15. Exam Mode & BCU Model Answer Generator

Specifically calibrated to the examination patterns of Bengaluru City University (BCU), Central Universities, and state university MSc programs:

- **2-Mark Mode:** Generates crisp 2-to-3 sentence definitions, balanced equations, or IUPAC names for Section A questions.
- **5-Mark Mode:** Formats answers into the standard 5-part university rubric: Principle, Balanced Reaction, Stepwise Mechanism with Intermediates, Synthetic Applications, and Summary.
- **10-Mark Mode:** Comprehensive long-form essays complete with orbital transition state diagrams, stereochemical proofs, retrosynthetic analysis, and industrial examples.

16. Academic Audit & Attendance Export System (PDF & Excel/CSV)

ChemBuddy provides full compliance documentation for college attendance committees, department heads, and scholarship audits:

16.1 Official Attendance PDF Report

- **Student Information Card:** Student Full Name, Register Number, University / College, and Current Semester.
- **Executive KPI Cards:** Overall Attendance Percentage (green $\geq 75\%$, red $< 75\%$), Sessions Count (Attended, Missed, Excused/OD), and Academic Standing (Safe skips allowed vs. consecutive classes required to reach 75%).
- **Coursework & Subject-Wise Performance Table:** Tabular breakdown showing Subject Name, Subject Code, Present, Absent, OD, Percentage, and individualized status prescriptions.
- **Official Verification Footer:** Certified footer with date and page numbers on every sheet.

16.2 Attendance Excel Spreadsheet (CSV Format)

- **Excel Compatible:** Encoded with UTF-8 BOM so all symbols and numbers open without formatting errors.
- **Subject Breakdown:** Columns for Attended, Missed, Excused, Total Counted, Percentage, Safe Skips (75%), Classes Needed (75%), Safe Skips (80%), and Classes Needed (80%).
- **Chronological Session Log:** Detailed historical record of every single marked class with Date, Subject Name, Status (Present/Absent/Excused/Postponed), and student notes.

16.3 Coursework & Topic Mastery Dossier

Exports an academic dossier detailing quiz accuracies, total questions answered, flashcards mastered, active study streaks, and competency ratings for each chemistry topic (Strong, Moderate, Needs Review).

17. Settings, Local Backup, Privacy & Offline Architecture

- **Notification Toggles:** Individual controls for Class Reminders, 8:30 AM Timetable Briefing, 8:00 PM Psychology Facts, and Exam Deadlines.
- **Instant Test Dispatch:** Tap "Send Test Notification" or "Send Psychology Fact" to verify notifications immediately.
- **Local JSON Export & Restore:** Export all profile records, attendance logs, timetable entries, and custom flashcards as an unencrypted portable JSON file. Restore seamlessly on any new device.
- **Zero Cloud Lock-in:** ChemBuddy never requires an account or cloud sign-in. Your data belongs exclusively to you.

18. Troubleshooting & Frequently Asked Questions (FAQ)

Q1: Why didn't my 8:30 AM timetable notification sound?

Answer: On Android 12+, ensure that "Alarms & Reminders" permission is granted in Android System Settings. Additionally, set battery usage for ChemBuddy to "Unrestricted" so Android's Doze mode does not defer morning alarms.

Q2: Should I keep my older app or install the new APK?

Answer: You must **uninstall / delete the older app** and install `ChemBuddy-v3.5.2-by-Prajwal-A-Kambar.apk`. The new APK contains the updated graph BFS/DFS SMILES generator, Gemini AI reaction prediction, reaction SVG export, and decluttered personalized home screen compiled into the native binary.

Q3: How does ChemBuddy calculate attendance if a class is cancelled?

Answer: Mark the class as *Postponed / Cancelled (/)*. ChemBuddy excludes that class from both the numerator and denominator, ensuring your percentage is never unfairly penalized.

Q4: Does Ask ChemBuddy work without internet?

Answer: The core features (attendance, 16 calculators, spectroscopy solver, pericyclic rules, flashcards, and 45+ offline reaction predictions) work 100% offline. Online Gemini AI mode connects over the internet to generate dynamic mechanisms for unlisted or custom molecules.

Q5: How do I export my attendance and reactions to Excel or PDF?

Answer: Navigate to the **Profile tab** or open any reaction in the **Reaction Mechanisms screen** and tap the **Export / Share icon** in the top-right corner. You can choose Academic PDF Dossier, Excel Spreadsheet (CSV/XLV), or standalone Vector SVG Diagram.

ChemBuddy v3.5.2 Visual Operating Manual

Engineered with ❤️ for MSc Chemistry Students worldwide by **Prajwal A Kambar**.